

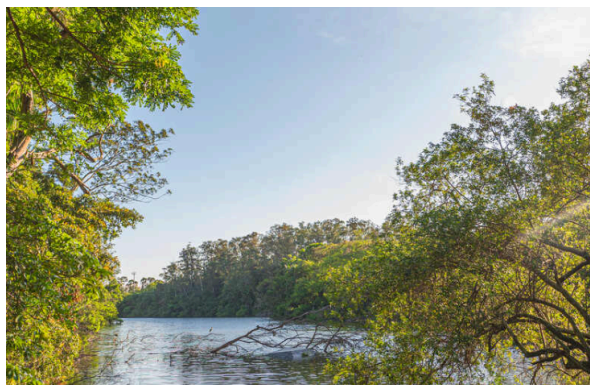
Chapter 20: Biodiversity

Chapter Concepts

- **Biodiversity** refers to the variation among organisms and ecological systems at all levels, including genetic, morphological, functional, and ecosystem differences.
- **Variation in the relative abundance of species** influences concepts of biodiversity.
 - Not all species are equally common; some are abundant, others rare.
 - Relative abundance affects ecosystem function and stability.
 - Indices and plots help quantify and compare abundance patterns.
- **Species richness** increases toward the equator and is influenced by latitude, environmental heterogeneity, and productivity.



- **Species richness** is highest in tropical regions and decreases toward the poles.
 - **Mountains** increase local diversity by creating varied habitats.
 - **Arid areas** reduce diversity due to harsh conditions.
 - This global pattern is fundamental for understanding biodiversity distribution.
- **Relative abundance** of species is important for understanding community structure. Rank-abundance plots show that most communities have a few **dominant species** and many rare ones.

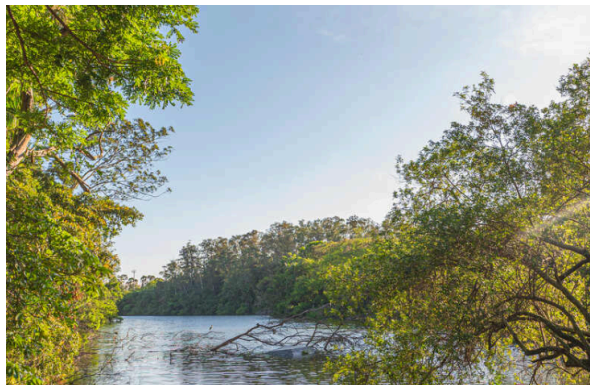


- **Rank-abundance plots** display the proportion of each species in a community.
 - Most communities have a few **dominant** and many **rare** species.
 - Abundance can be measured by number, biomass, or cover.
 - Understanding abundance helps reveal community stability and function.

- **Species–area relationship:** The number of species increases with the area sampled, following the equation $S = cc \cdot A^z$.



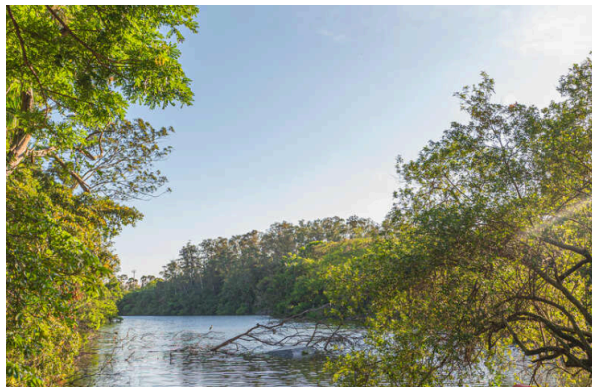
- Larger areas support more species due to increased habitat diversity.
 - The relationship is consistent across different taxa and regions.
 - The slope z reflects how quickly richness increases with area.
 - This principle is used in conservation to estimate species loss from habitat reduction.
- The **slope** of the species–area relationship is influenced by local sampling, habitat diversity, and evolutionary history.



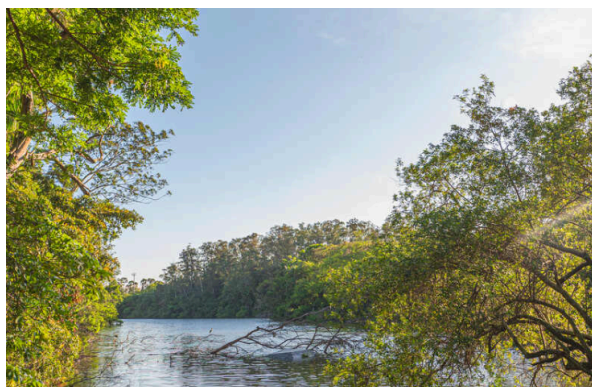
- At small scales, sampling more individuals increases species count rapidly.
 - At regional scales, new habitat types are included, increasing richness.
 - At global scales, evolutionary history and continental separation further increase diversity.
 - The curve's shape reveals the dominant processes at each scale.
- **Large-scale patterns of diversity** reflect latitude, environmental heterogeneity, and productivity.
 - Diversity increases toward the equator and in complex, productive habitats.
 - Both regional and local factors shape these patterns.
- **Latitudinal gradients:** Species richness increases toward the tropics.



- Mammal species richness rises from north to south in North America.
- Western mountains have higher diversity due to topographic heterogeneity.
- Fewer species occur on peninsulas due to dispersal limitations.
- These patterns reflect both climate and geographic barriers.
- **Environmental heterogeneity** and **topography** increase diversity.



- Trees and amphibians are more diverse in moist eastern regions.
- Reptile diversity is highest in warm southern areas.
- Environmental gradients (moisture, temperature) shape species distributions.
- Different taxa respond to different environmental factors.
- **Habitat diversity** and **structural complexity** increase species richness.

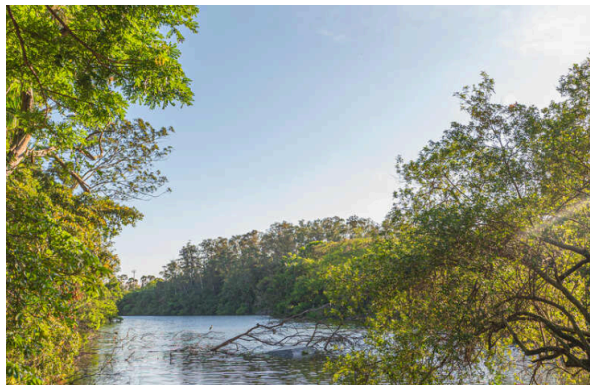


- Bird species richness increases with both productivity and habitat complexity.
- Forests support more species than grasslands or marshes.
- Simple habitats have fewer ecological niches.
- Structural complexity allows for more specialized species.

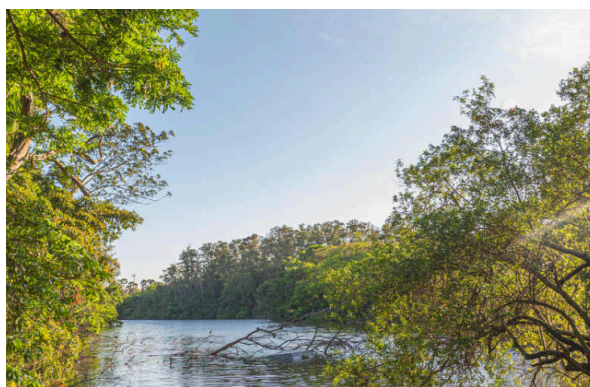
- **Habitat productivity** and **structural complexity** both contribute to higher diversity.



- Highly productive marshes may have low diversity if structurally simple.
 - Deserts, though less productive, can have higher diversity due to varied structure.
 - Productivity and structure interact to determine community richness.
 - Management should consider both factors for biodiversity conservation.
- **Foliage height diversity** is positively correlated with bird species diversity.



- More layers in vegetation support more bird species.
 - Vertical structure creates diverse foraging and nesting opportunities.
 - This relationship is observed across different forest types.
 - Habitat management can enhance bird diversity by increasing foliage layers.
- **Productivity** and **energy input** (e.g., solar energy, precipitation) are correlated with species richness.

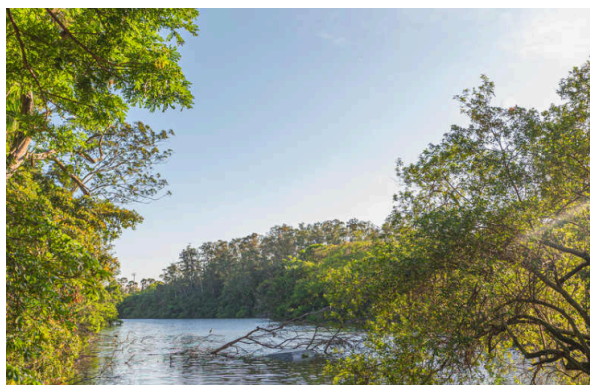


- In temperate regions, energy input (temperature, sunlight) predicts richness.

- In the tropics, water availability is more important.
- Both factors interact to shape global diversity patterns.
- Understanding these drivers helps predict responses to climate change.
- **Potential evapotranspiration (PET)** is a predictor of species richness, especially in vertebrates.



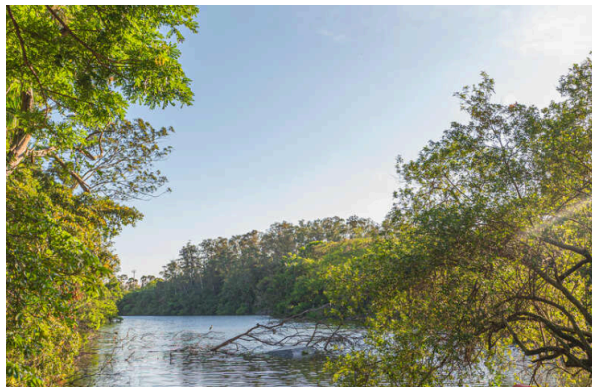
- Species richness increases with PET up to a threshold, then levels off.
- PET integrates temperature and solar radiation.
- Above certain PET values, other factors (e.g., aridity) limit diversity.
- PET is a useful index for comparing regions.
- **Diversity has both regional and local components.**
 - **Alpha diversity:** local richness.
 - **Gamma diversity:** regional richness.
 - **Beta diversity:** turnover between habitats.
- **Beta diversity:** Turnover of species between habitats, measured by indices such as **Sørensen similarity** and **Jaccard similarity**.



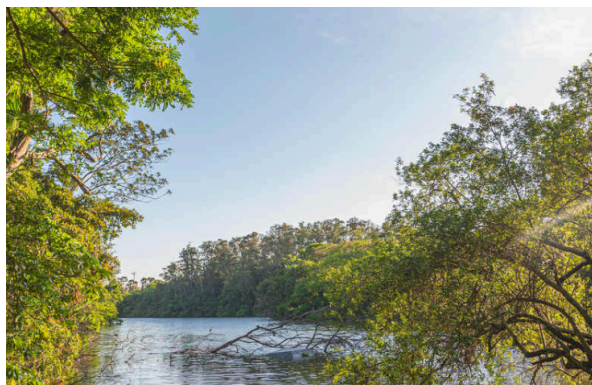
- Sørensen similarity decreases with distance, indicating species turnover.
- Eastern Asia shows higher turnover than North America.
- Turnover is greater along steep climate gradients.
- Beta diversity reflects both environmental and historical factors.



- Jaccard similarity decreases more steeply in southern regions.
- Climate and distance both affect species turnover.
- Beta diversity is key for understanding regional patterns.
- Conservation planning must account for high-turnover areas.
- **Species sorting** is determined by environmental filters and species interactions.



- Physical factors (e.g., wave exposure) determine which species dominate.
- Species are sorted into habitats based on tolerance and competition.
- Community composition changes with environmental gradients.
- Sorting explains local differences within a region.



- Experimental microcosms show how environmental filters operate.
- Only species adapted to specific conditions persist.
- Filters include water level, soil fertility, and competition.
- Species pools are reduced by multiple filtering steps.
- **Species sorting** means local communities are subsets of the regional species pool.

- Membership is determined by both adaptations and interactions.
- Environmental filters and biotic interactions act together.
- **Ecological release** occurs when species expand their niche in less diverse communities.



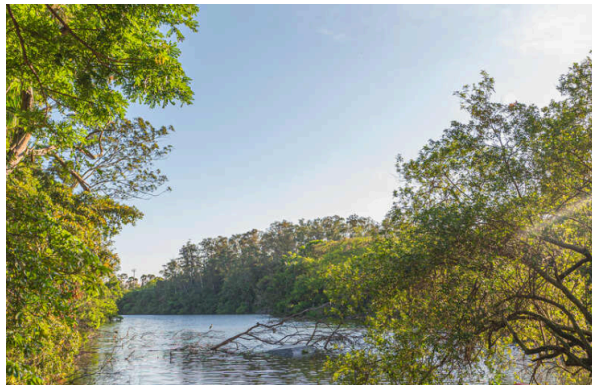
- On islands or depauperate regions, species occupy more habitats.
- Local abundance and niche breadth increase as regional diversity decreases.
- Ecological release demonstrates the role of competition in limiting niches.
- Fewer competitors allow species to exploit more resources.
- **Diversity can be understood in terms of niche relationships.**
 - Niche breadth, overlap, and specialization determine coexistence.
 - More niche space allows more species.
- **Niche relationships** and **competition** structure communities; more diverse communities have more niche space.



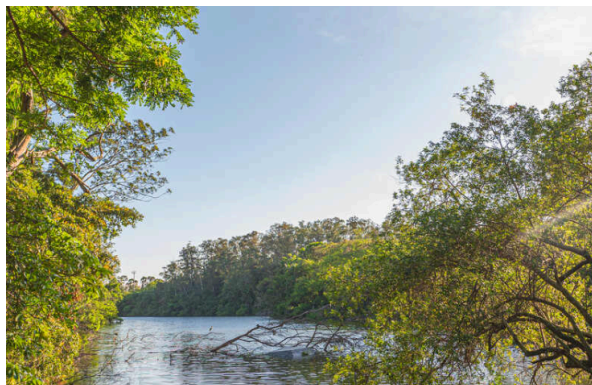
- Morphological diversity reflects ecological roles.
- Tropical bat communities occupy more niche space than temperate ones.
- Niche overlap and specialization determine coexistence.
- Functional diversity increases with species richness.



- Fish communities show increasing ecological roles downstream.
- Larger, more complex habitats support more feeding types.
- Community structure reflects both habitat and resource diversity.
- Niche diversity is linked to environmental gradients.
- **Equilibrium theories of diversity** balance factors that add and remove species.
 - Immigration, speciation, and extinction interact to set diversity.
 - Island biogeography is a key model.
- **Equilibrium theory of island biogeography** explains species richness as a balance between immigration and extinction.

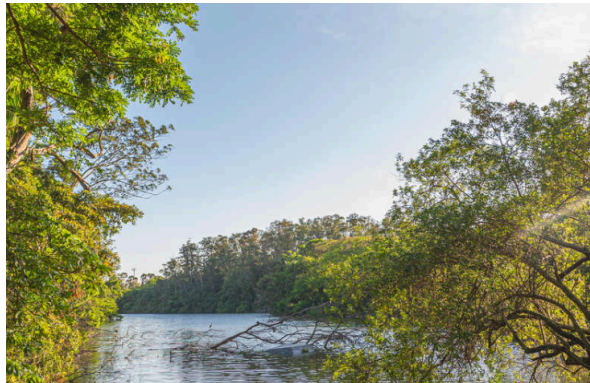


- Immigration rate decreases as more species colonize an island.
- Extinction rate increases with more species present.
- Equilibrium is reached where immigration equals extinction.
- The model predicts steady-state species richness.

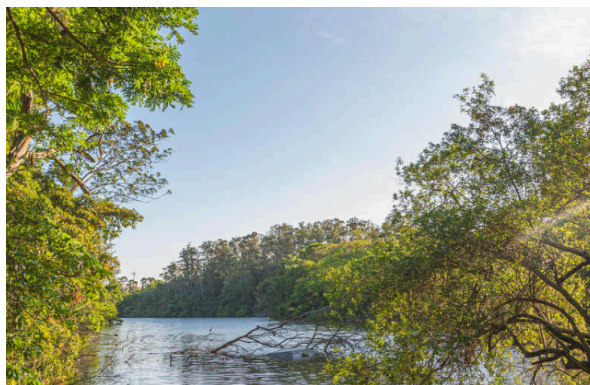


- Small islands have higher extinction rates and support fewer species.

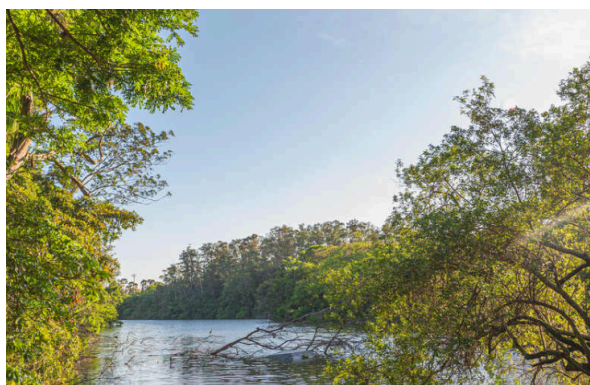
- Large islands are better targets for colonists and have lower extinction.
- Island size is a key determinant of diversity.
- Conservation of large habitats is critical for maintaining richness.



- Islands closer to the mainland have higher immigration rates.
- Distance from source populations limits colonization.
- Near islands support more species than far islands.
- Connectivity is important for sustaining island diversity.

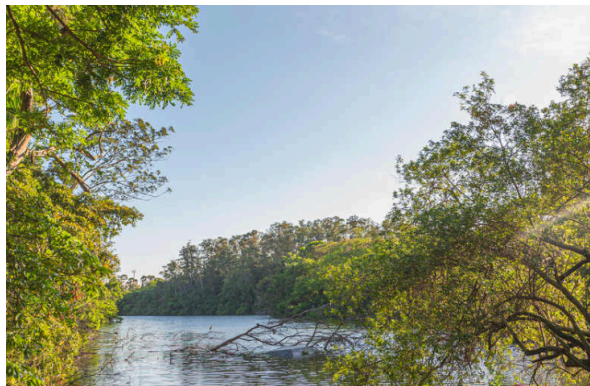


- Experimental defaunation shows rapid recolonization of near islands.
- Species richness returns to equilibrium after disturbance.
- Recovery is faster for islands closer to colonist sources.
- The experiment supports equilibrium theory predictions.

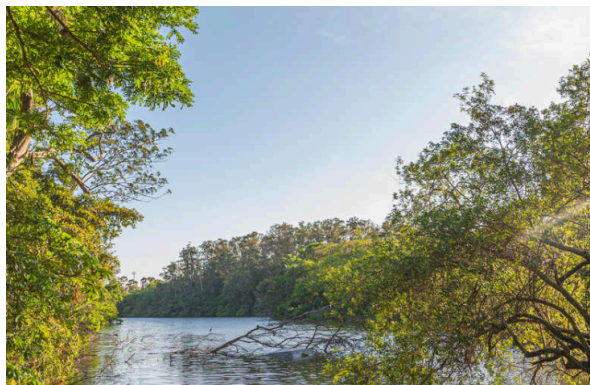


- On continents, speciation adds to regional diversity.
- Equilibrium is influenced by both immigration and speciation.
- Large landmasses support evolutionary radiations.

- Regional processes shape continental diversity patterns.
- **Explanations for high tree species richness in the tropics** focus on forest dynamics.
 - Environmental heterogeneity, disturbance, pest pressure, and slow exclusion all contribute.
 - No single factor fully explains the latitudinal gradient.
- **Herbivores** and **pathogens** (pest pressure) can prevent competitive exclusion and maintain diversity.



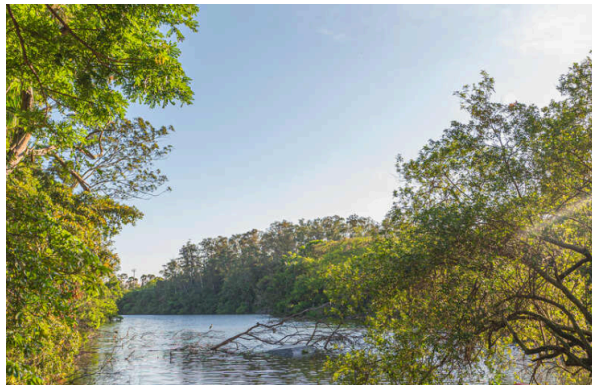
- Seedling survival is highest away from parent trees.
- Herbivores and pathogens concentrate near adults, reducing local recruitment.
- Density-dependent mortality promotes coexistence.
- The pest pressure hypothesis explains high tropical tree diversity.



- Fungicide treatment increases seedling survival in dense plots.
- Pathogens are a major cause of seedling mortality.
- Reducing pathogen pressure alters community structure.
- Management of pests can impact diversity outcomes.
- **Pest pressure hypothesis:** Herbivores and pathogens promote diversity by limiting dominant species.
 - Density-dependent mortality favors rare species.
 - Pathogens and herbivores prevent competitive exclusion.
- **Random ecological drift** (neutral theory) suggests that stochastic processes can maintain diversity over long timescales.



- All individuals have equal probabilities of dying and reproducing.
- Community composition changes by chance, not niche differences.
- Drift is slow and requires large populations for persistence.
- Neutral theory highlights the role of randomness in diversity.
- **Random ecological drift** (neutral theory) complements niche-based explanations.
 - Drift is slow and requires large populations.
 - Neutral theory is most relevant at local scales or in very diverse communities.
- **Species accumulation curves** and **rarefaction curves** are used to estimate species richness from samples.



- Accumulation curves show how richness increases with sampling effort.
- Rarefaction standardizes richness estimates for fair comparison.
- Both methods help compare diversity across habitats or treatments.
- Proper sampling is essential for accurate biodiversity assessment.